W2 - OS Lab Work

1WA24CS241

1. Algorithm FCFS

Input: List of processes with (ProcessID, ArrivalTime, BurstTime)

Output: CompletionTime, TurnaroundTime, WaitingTime for each process

Code >>>

```
#include <stdio.h>

int main() {
   int n;
   int at[20];
   int bt[20];
   int wt[20];
   int tat[20];
   float twt = 0.0;
   float ttat = 0.0;
   float awt;
   float atat;

   printf("Stawan H. Ruiwale, 1WA24CS241\n\n");

   printf("Enter the num of process : ");
   scanf("%d", &n);

   printf("Enter the arrival time for each process: ");
   for (int i = 0; i < n; i++) {
      printf("Process %d Arrival time : ", i + 1);
      scanf("%d", &at[i]);
   }

   printf("Enter the burst time for each process: ");
   for (int i = 0; i < n; i++) {
      printf("Process %d Burst time : ", i + 1);
      scanf("%d", &bt[i]);
   }

    // Calc wt and tat
   wt[0] = 0; //doesn't wait
   tat[0] = bt[0]; //tat for the 1st process is its bt
   for (int i = 1; i < n; i++) {
```

```
        //wt of current process = wt of previous process + bt of previous process
        wt[i] = wt[i - 1] + bt[i - 1];
        // tat = wt + bt
        tat[i] = wt[i] + bt[i];
    }

    // Calc total and avg times
    for (int i = 0; i < n; i++) {
        twt += wt[i];
        ttat += tat[i];
    }
    awt = twt / n;
    atat = ttat / n;

    // Display results
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t\t%d\t\t%d\n", i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time: %.2f\n", awt);
    printf("Average Turnaround Time: %.2f\n", atat);

    return 0;
}
```

Output

>>>

```
"C:\Users\Admin\Documents\

Stawan H. Ruiwale, 1WA24CS241

Enter the num of process : 4
Enter the arrival time for each process: Process 1 Arrival time : 0
Process 2 Arrival time : 0
Process 3 Arrival time : 0
Process 4 Arrival time : 0
Enter the burst time for each process: Process 1 Burst time : 7
Process 2 Burst time : 3
Process 3 Burst time : 4
Process 4 Burst time : 6

Process Burst Time      Waiting Time    Turnaround Time
P1      7               0               7
P2      3               7               10
P3      4               10              14
P4      6               14              20

Average Waiting Time: 7.75
Average Turnaround Time: 12.75

Process returned 0 (0x0)   execution time : 29.151 s
Press any key to continue.
```

2. Algorithm SJF_NonPreemptive :

Code

>>>

```
#include <stdio.h>

int main() {
   int n, i, j;
   int pid[20], at[20], bt[20], ct[20], tat[20], wt[20];
   int completed[20] = {0};
   int current_time = 0, completed_count = 0;
   int min_bt, index;

    printf("Stawan H. Ruiwale, 1WA24CS241\n\n");


   printf("Enter number of processes: ");
   scanf("%d", &n);

   for(i = 0; i < n; i++) {
      printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
      scanf("%d %d", &at[i], &bt[i]);
      pid[i] = i + 1;
   }

   while(completed_count < n) {
      min_bt = 9999;
      index = -1;

      for(i = 0; i < n; i++) {
         if(at[i] <= current_time && completed[i] == 0) {
            if(bt[i] < min_bt) {
               min_bt = bt[i];
               index = i;
            }
         }
      }

      if(index == -1) {
         current_time++;
      } else {
         ct[index] = current_time + bt[index];
         tat[index] = ct[index] - at[index];
         wt[index] = tat[index] - bt[index];

         current_time = ct[index];
```

```
        completed[index] = 1;
        completed_count++;
      }
    }

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
    for(i = 0; i < n; i++) {
      printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
          pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
    }

    return 0;
}
```

Output

>>>

```
"C:\Users\Admin\Documents\

Stawan H. Ruiwale, 1WA24CS241

Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 0 7
Enter Arrival Time and Burst Time for P2: 8 3
Enter Arrival Time and Burst Time for P3: 3 4
Enter Arrival Time and Burst Time for P4: 5 6

PID     AT      BT      CT      TAT     WT
P1      0       7       7       7       0
P2      8       3       14      6       3
P3      3       4       11      8       4
P4      5       6       20      15      9

Process returned 0 (0x0)   execution time : 30.220 s
Press any key to continue.
```

3. Algorithm SJF Preemptive :

Code

>>>

```
#include <stdio.h>

int main() {
   int n, i;
   int pid[20], at[20], bt[20], rt[20];
   int ct[20], tat[20], wt[20];
   int completed = 0, current_time = 0;
   int min_index, min_rt;

   printf("Stawan H. Ruiwale, 1WA24CS241\n\n");

   printf("Enter number of processes: ");
   scanf("%d", &n);

   for(i = 0; i < n; i++) {
      printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
      scanf("%d %d", &at[i], &bt[i]);

      pid[i] = i + 1;
      rt[i] = bt[i];
   }

   while(completed < n) {
      min_rt = 9999;
      min_index = -1;

      for(i = 0; i < n; i++) {
         if(at[i] <= current_time && rt[i] > 0) {
            if(rt[i] < min_rt) {
               min_rt = rt[i];
               min_index = i;
            }
         }
      }

      if(min_index == -1) {
         current_time++;
      } else {
         rt[min_index]--;
         current_time++;

         if(rt[min_index] == 0) {
```

```
            completed++;

            ct[min_index] = current_time;
            tat[min_index] = ct[min_index] - at[min_index];
            wt[min_index] = tat[min_index] - bt[min_index];
        }
    }
  }

  printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
  for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
          pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
  }

  return 0;
}
```

Output

>>>

```
"C:\Users\Admin\Documents\

Stawan H. Ruiwale, 1WA24CS241

Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 0 7
Enter Arrival Time and Burst Time for P2: 8 3
Enter Arrival Time and Burst Time for P3: 3 2
Enter Arrival Time and Burst Time for P4: 5 6

PID     AT      BT      CT      TAT     WT
P1      0       7       9       9       2
P2      8       3       12      4       1
P3      3       2       5       2       0
P4      5       6       18      13      7

Process returned 0 (0x0)   execution time : 32.413 s
Press any key to continue.
```